

RESEARCH INTERESTS

Trustworthy Machine Learning; Security of Retrieval-Augmented and Multi-Tool LLM Systems; Out-of-Distribution / Novelty Detection; Clustering and Anomaly Detection.

EDUCATION

Shanghai Normal University 2022.09 - 2025.06 | Shanghai, China

Master of Engineering in Computer Science and Technology
Supervisor: Prof. Yan Ma
Master thesis: "Research on a Split-Merge Clustering Algorithm Based on Minimum Spanning Forests"
Award: Outstanding Graduate Award, Shanghai Normal University, 2025 (top 5%)
Major courses: Digital Image Processing (90), Digital Signal Processing (90), Computer Network and Simulation (94)

Nanjing University of Aeronautics and Astronautics Jincheng College 2017.09 - 2021.06 | Jiangsu, China

Bachelor of Engineering in Transportation
Major courses: Linear Algebra (93), Advanced Mathematics (98), Probability Theory and Mathematical Statistics (100)

PUBLICATIONS

- [1] **Haoyu Zhai**, Jie Yang, Haotao Guo, Bin Wang, Yan Ma. "Density-increment and cut-edge optimized clustering via minimum spanning forest." **Neurocomputing**, 2026. [First author | JCR Q1]. DOI: <https://doi.org/10.1016/j.neucom.2026.132957>. [code]
- [2] Zexuan Fei, **Haoyu Zhai**, Jie Yang, Bin Wang, Yan Ma. "Discovering generalized clusters with adaptive mixture density-based clustering." **Knowledge-Based Systems**, 2025. [Co-first author | JCR Q1]. DOI: <https://doi.org/10.1016/j.knosys.2025.113250>. [code]
- [3] **Haoyu Zhai**, Zexuan Fei, Yan Ma. "A Novel Outlier Detection Algorithm Based on Symmetry and Distance Ratio." **International Conference on Pattern Recognition**, 2024. [First author]. DOI: https://doi.org/10.1007/978-3-031-78192-6_22. [code]

RESEARCH EXPERIENCE

360 Group, Beijing -- AI Algorithm Engineer 2025.07 - Present

- Embedding-Based Code / Malware Similarity Retrieval (Milvus)*
Role: Sole developer; validated the route before later APT-team scale-up.
- Designed and implemented an embedding-based code-similarity retrieval pipeline for malware analysis: code-file preprocessing, chunk splitting, embedding generation, Milvus indexing, approximate nearest-neighbor (ANN) search, and file-level aggregation.
 - Used GraphCodeBERT (later Qwen3-Embedding-0.6B) to embed code chunks; built HNSW + COSINE ANN indexes in Milvus; aggregated chunk-level hits to file level via Reciprocal Rank Fusion (RRF) for code deduplication, same-origin sample association, and partial reused-malware-snippet detection.
 - Validated the approach on around 17.6K files / 3.5M chunk embeddings; an upgraded model and index design extended it to around 40M chunk embeddings with under 10 ms average retrieval latency, over 95% Top-1 hit rate at effective-subset ratio ≥ 0.4 , and around 3.2x embedding-throughput speedup.
- Multi-Tool LLM Agent System for Malware and Security Analysis* – lead developer
- Built a multi-tool LLM agent platform for malware/security triage, orchestrating Python, Node.js, shell execution, file operations, binary/PE inspection, and Office/PDF analysis in Docker sandboxes.
 - Implemented batch multi-agent dispatch and a FastAPI WebUI for file upload, task submission, live status tracking, checklist progress, and Markdown report rendering.
 - Diagnosed recurring failures in the agent pipeline, tracing them primarily to base-model capability limits, and improved end-to-end reliability through model selection and prompt design.

NSFC Project: Graph-Based Clustering and Image Recognition 2022.09 - 2025.06 | Shanghai Normal University

- Role: Algorithm designer / core contributor
- Designed graph-based clustering and outlier-detection algorithms, including symmetry-distance outlier scoring and a minimum-spanning-forest split/merge clustering framework.
 - Evaluated SMMSF on 12 two-dimensional and 15 real-world datasets against 9 algorithms; used AC, PR, RE, and F1 metrics and achieved best results on 13 real-world datasets.
 - Evaluated the outlier detector on 4 synthetic and 16 UCI datasets against 5 mainstream detectors; used denoising-before-clustering experiments to verify clustering gains.
 - Contributed to related patent/application materials from the project under the supervision of senior collaborators.

SKILLS

- Research:** Trustworthy ML, LLM agent reliability/security, clustering, anomaly detection, OOD/novelty detection, computer vision.
- Engineering:** Python, C/C++, Matlab, PyTorch, Scikit-learn, OpenCV, Docker, FastAPI, Linux, Git, Milvus, LaTeX/Typst.
- Languages:** Mandarin Chinese (native), English (working proficiency).